CS & IT ENGINEERING

Computer Network

IPv4 Address

Lecture No. – 08

By - Abhishek Sir

Recap of Previous Lecture

Topic Routing Table

Topics to be Covered

- **Topic** Forwarding
- **Topic** ARP

ABOUT ME

Hello, I'm Abhishek

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[MCQ] [GATE-CS-2004] [2 Mark]

#Q. The routing table of a router shown below :

Destination	Subnet Mask	Interface
128 . 75 . 43 . 0	255 . 255 . 255 . 0	Eth0
128 . 75 . 43 . 0	255 . 255 . 255 . 128	Eth1
192 . 12 . 17 . 5	255 . 255 . 255 . 255	Eth3
Default		Eth2

On which interfaces will the router forward packets addressed to destinations 128 . 75 . 43 . 16 and 192 . 12 . 17 . 10 respectively ?

A. Eth1 and Eth2

B. Eth0 and Eth2

C. Eth0 and Eth3

D. Eth1 and Eth3

Ans: A

128. 75 . 43.16
255.255.255.255
----
128. 75 . 43. 16 ✗

128. 75 . 43.16
255.255.255.128
----
128. 75 . 43. 0 ✓ Eth1

192. 12 . 17. 10
255.255.255.255
----
192. 12 . 17. $\boxed{10}$ ✗

Default Eth2

[NAT] [GATE-2023][2 Mark]

#Q. The forwarding table of a router is shown below.

Subnet Number	Subnet Mask	Interface ID
200.150.0.0	255.255.0.0	1
200.150.64.0	255.255.224.0	2
200.150.68.0	255.255.255.0	3 ✓
200.150.68.$\boxed{64}$	255.255.255.240	4 ×
Default		0

A packet addressed to a destination address 200.150.68.118 arrives at the router. It will be forwarded to the interface with ID ________.

$\boxed{\text{Ans} = 3}$

200.150. 68.118

255.255.255.240

200.150. 68.112 ✗

$118 \rightarrow 01110110$

$240 \rightarrow 11110000$

$112 \leftarrow 01110000$

200.150. 68.118

255.255.255. 0

200.150. 68. 0 ✓

[MCQ] [GATE-2025][2 Mark]

#Q. A packet with the destination IP address 145.36.109.70 arrives at a router whose routing table is shown. Which interface will the packet be forwarded to?

Subnet	Subnet Mask (in CIDR notation)	Interface
145.36.0.0	/16	E1
145.36.128.0	/17	E2
145.36.64.0	/18	E3 ✓
145.36.225.0	/24	E4 ×
0.0.0.0 Default	/0 --	E5

(A) E3 ✓

(B) E1

(C) E2

(D) E5

$\boxed{\text{Ans: A}}$

145. 36.109. 70
255.255.255. 0
----
145. 36. $\boxed{109}$. 0

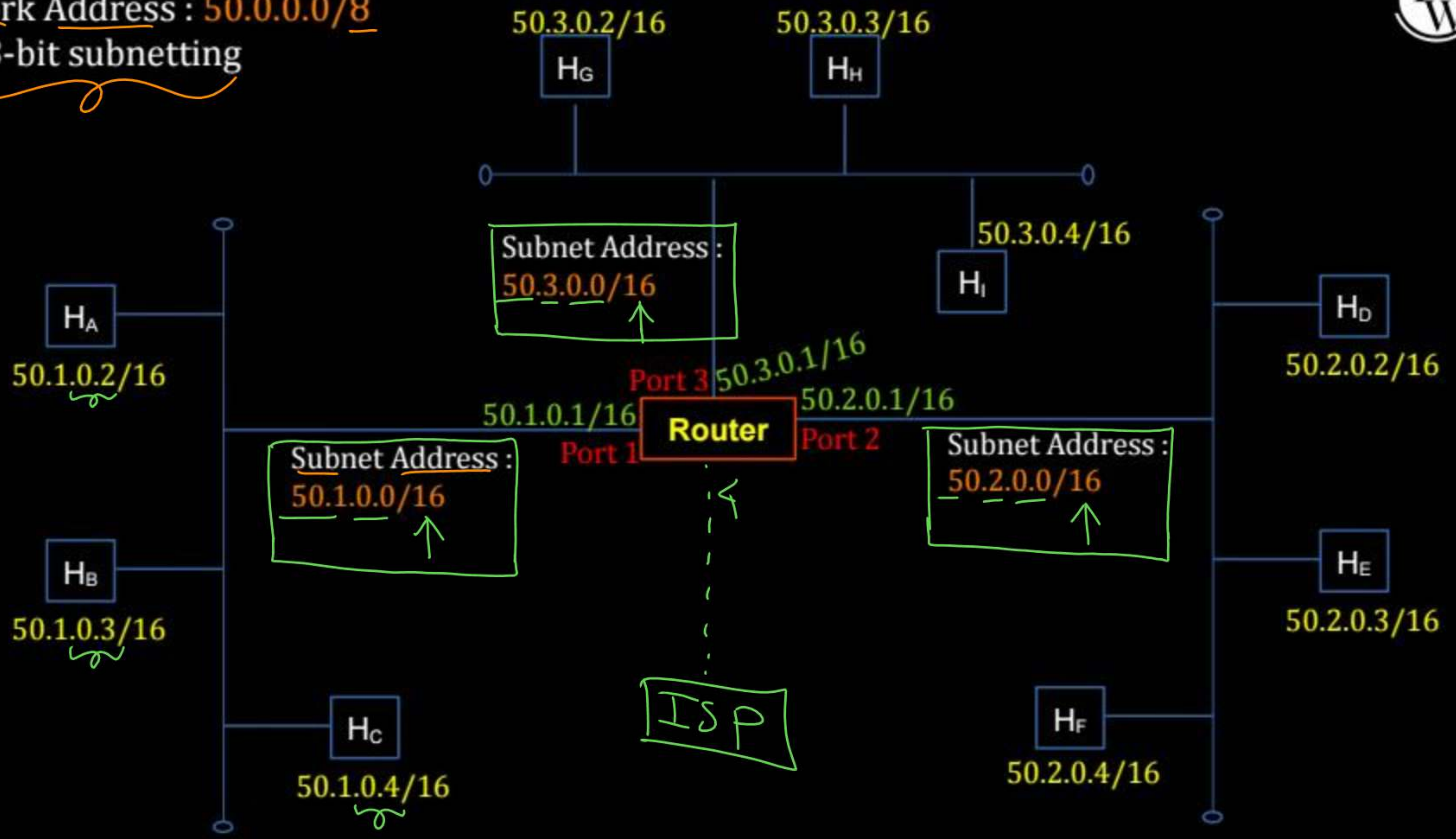
×

145. 36.109. 70
255.255.192. 0
----
145. 36. 64. 0

109 → 01101101
192 → 11000000

64 ← 01000000

Network Address : 50.0.0.0/8
With 8-bit subnetting



Network Address : 50.0.0.0/8
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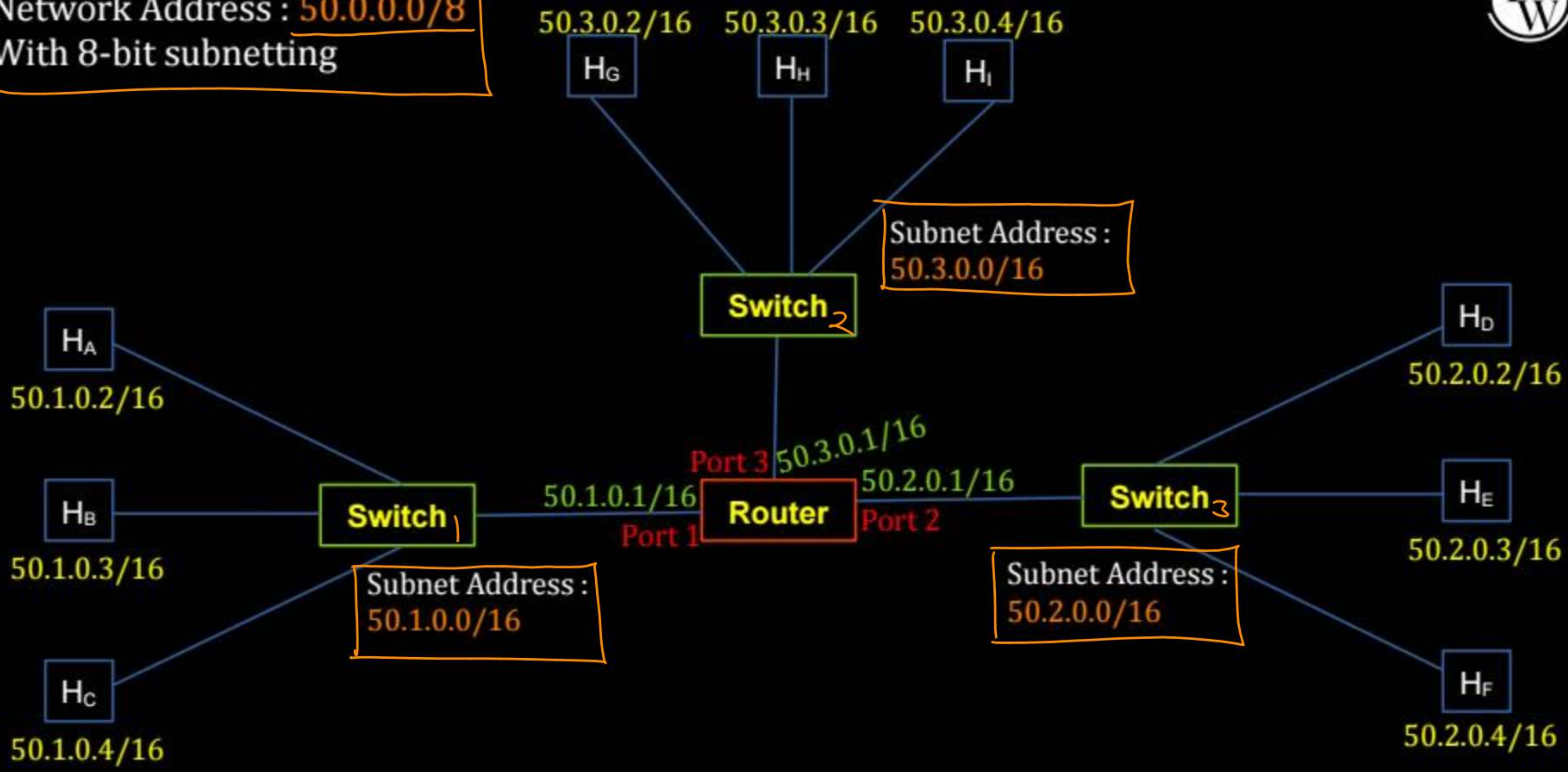


Host A, IPv4 Configuration :-

IPv4 Address : 50 . 1 . 0 . 2

Subnet Mask : 16 or 255 . 255 . 0 . 0

Default Gateway : 50 . 1 . 0 . 1

Host A

50.1.0.2
255.255.0.0
50.1.0.0

Router forwarding table

Subnet Address	Interface ID	Next Hop
50 . 1 . 0 . 0 / 16	1	On Link
50 . 2 . 0 . 0 / 16	2	On Link
50 . 3 . 0 . 0 / 16	3	On Link

CASE I :

IP Packet

Source IP Address	: 50 . 1 . 0 . 2	HA
Destination IP Address	: 50 . 1 . 0 . 3	HB

Host A

50.1.0.3
255.255.0.0
50.1.0.0

→ Host A (source host) finds destination host IP (Host B) belongs to same subnet

→ Host A uses ARP Protocol to find MAC Address of destination host

→ Host A send frame to Host B
[The frame encapsulates the IP datagram]

Source MAC Address	: Host A MAC Address
Destination MAC Address	: Host B MAC Address

Frame (IP Packet)

Node to Node

CASE II : (Part-01)

IP Packet

Host A
IP setting
Mask → 16

Source IP Address	:	50 . 1 . 0 . 2 $\boxed{H_A}$
Destination IP Address	:	50 . 2 . 0 . 2 $\boxed{H_D}$

→ Host A (source host) finds destination host IP (Host D) belongs to different subnet

→ Host A uses ARP Protocol to find MAC Address of default gateway [50 . 1 . 0 . 1]

→ Host A send frame to Router
[The frame encapsulates the IP datagram]

Source MAC Address	:	Host A MAC Address
Destination MAC Address	:	Router MAC Address

Frame [IP Packet]

Node to Node

CASE II : (Part-02)

IP Packet

Source IP Address	:	50 . 1 . 0 . 2
Destination IP Address	:	50 . 2 . 0 . 2

H_A

H_D

→ Router receives the IP datagram

→ Finds destination host IP (Host D) belongs to subnet connected via port 2

→ Router uses ARP Protocol to find MAC Address of destination host

→ Router send frame to Host D
[The frame encapsulates the IP datagram]

Source MAC Address	:	Router MAC Address
Destination MAC Address	:	Host D MAC Address

Frame_2[IP Packet]

CASE III :

Source IP Address	:	50 . 1 . 0 . 2
Destination IP Address	:	50 . 1 . 255 . 255

HA

/16

→ Host A (source host) finds destination ~~host~~ IP belongs to same subnet

→ Host A uses ARP, ARP give broadcast MAC Address

→ Host A broadcast the frame into same subnet
[The frame encapsulates the IP datagram]

Source MAC Address	:	Host A MAC Address
Destination MAC Address	:	Broadcast MAC Address [FF:FF:FF:FF:FF:FF]

CASE IV :

Source IP Address : 50 . 1 . 0 . 2 H_A

Destination IP Address : 255 . 255 . 255 . 255

→ Host A (source host) finds destination ~~host~~ IP is limited broadcast

→ Host A uses ARP, ARP give broadcast MAC Address

→ Host A broadcast the frame into same subnet
[The frame encapsulates the IP datagram]

Source MAC Address : Host A MAC Address

Destination MAC Address : Broadcast MAC Address
[FF:FF:FF:FF:FF:FF]

[MCQ] H.W. IIT-KGP [GATE-CS-2006] [2 Mark]

#Q. Two computers C1 and C2 are configured as follows. C1 has IP address "203 . 197 . 2 . 53" and netmask "255 . 255 . 128 . 0". C2 has IP address "203 . 197 . 75 . 201" and netmask "255 . 255 . 192 . 0". Which one of the following statements is true ?

A. C1 and C2 both assume they are on the same network

B. C2 assumes C1 is on same network, but C1 assumes C2 is on a different network

C. C1 assumes C2 is on same network, but C2 assumes C1 is on a different network

D. C1 and C2 both assume they are on different networks

[MCQ] H.W. [GATE-IT-2008] [2 Mark]

#Q. Host X has IP address 192 . 168 . 1 . 97 and is connected through two routers R1 and R2 to another host Y with IP address 192 . 168 . 1 . 80, Router R1 has IP addresses 192 . 168 . 1 . 135 and 192 . 168 . 1 . 110, R2 has IP addresses 192 . 168 . 1 . 67 and 192 . 168 . 1 . 155, the netmask used in the network is 255 . 255 . 255 . 224;

Which IP address should X configure its gateway as?

(A) 192 . 168 . 1 . 67

(B) 192 . 168 . 1 . 110

(C) 192 . 168 . 1 . 135

(D) 192 . 168 . 1 . 155

[MCQ] H.W. [GATE-IT-2008] [2 Mark]

#Q. Given the information in previous question, how many distinct subnets are guaranteed to already exist in the network?

A 1

B 2

C 3

D 6

[MSQ] IISC, H.W. [GATE-2024][1 Mark]

#Q. Node X has a TCP connection open to node Y. The packets from X to Y go through an intermediate IP router R. Ethernet switch S is the first switch on the network path between X and R. Consider a packet sent from X to Y over this connection.

Which of the following statements is/are TRUE about the destination IP and MAC addresses on this packet at the time it leaves X?

A. The destination IP address is the IP address of R

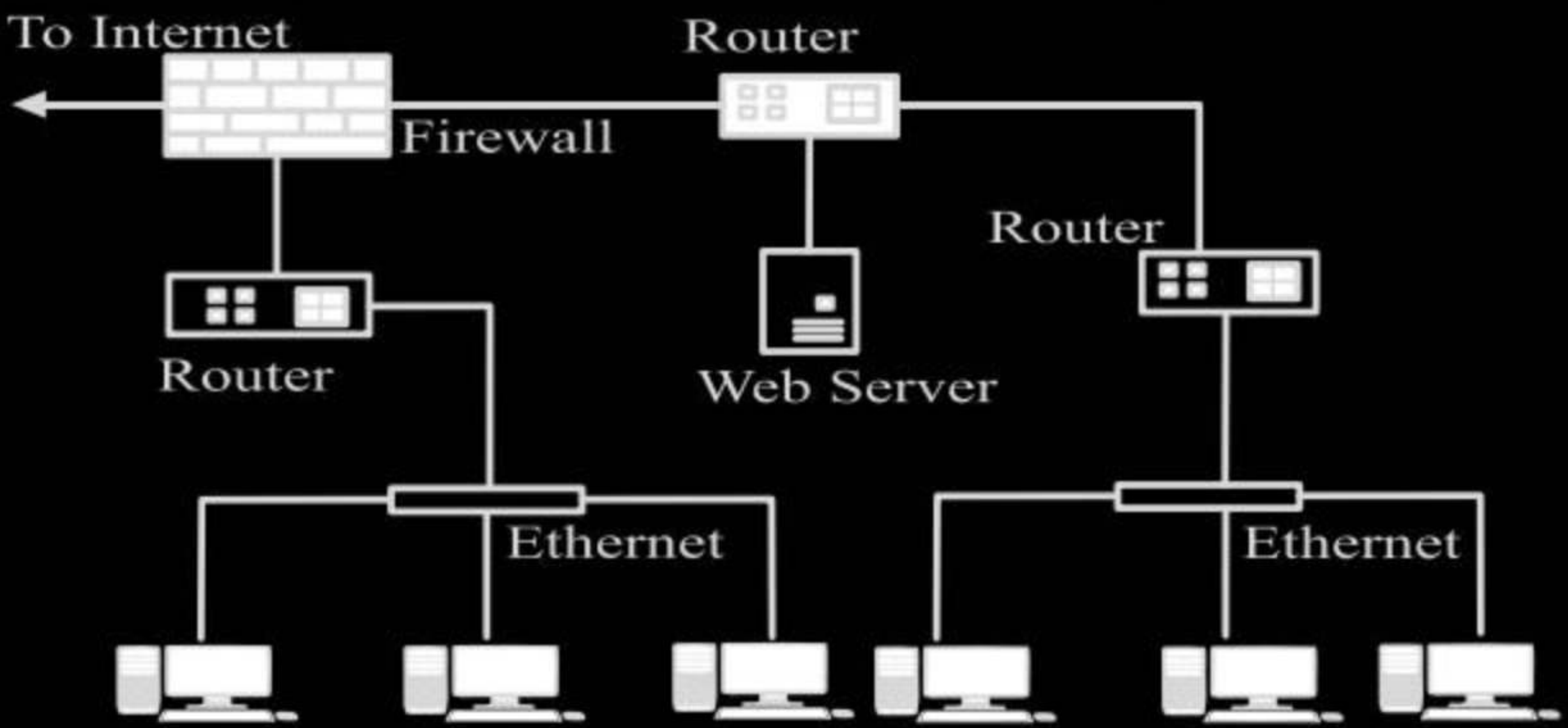
B. The destination IP address is the IP address of Y

C. The destination MAC address is the MAC address of S

D. The destination MAC address is the MAC address of Y

[MCQ] IIT-KGP, H.W. [GATE-2022][1 Mark]

#Q. Consider an enterprise network with two Ethernet segments, a web server and a firewall, connected via three routers as shown below.



What is the number of subnets inside the enterprise network?

(A) 3

(B) 12

(C) 6

(D) 8

Topic : ARP

- → ARP : Address Resolution Protocol
- → Data Link Layer Protocol /Network Layer Protocol
- → Used to find MAC address of device from IP address
- → Map logical address into corresponding physical address

Network Layer

$$
\begin{tikzcd}
\text{IP} \arrow[d, "\text{(IP Datagram)}"'] \arrow[r, "\text{Target IP Add.}"] & \text{ARP} \arrow[dl, "\text{MAC Address}"] \\
\text{Data Link Layer} &
\end{tikzcd}
$$

[MCQ] [GATE-CS-2005] [1 Mark]

#Q. The address resolution protocol (ARP) is used for :

~~A~~ Finding the IP address from the DNS

~~B~~ Finding the IP address of the default gateway

~~C~~ Finding the IP address that corresponds to a MAC address $\boxed{\text{RARP}}$

D Finding the MAC address that corresponds to an IP address $\boxed{\text{ARP}}$

$\boxed{\text{Ans: D}}$

[MCQ] IIT-R [GATE-2025] [1 Mark]

#Q. Consider the following statements :

(i) Address Resolution Protocol (ARP) provides a mapping from an IP address to the corresponding hardware (link-layer) address. TRUE

(ii) A single TCP segment from a sender S to a receiver R cannot carry both data from S to R and acknowledgement for a segment from R to S.

Which ONE of the following is CORRECT?

i → T/F

(A) Both (i) and (ii) are TRUE

(B) (i) is TRUE and (ii) is FALSE

(C) (i) is FALSE and (ii) is TRUE

(D) Both (i) and (ii) are FALSE

Topic : ARP Packet

→ ARP Packet always encapsulated into payload field of "Frame"

→ ARP Query (Request) always broadcast
[Destination MAC address = FF:FF:FF:FF:FF:FF (All 48 bits are one)
Broadcast MAC address]

→ ARP Reply always unicast
[Destination MAC address = MAC address of Host who raised ARP query]

[MCQ] (IT-B) [GATE-2021][1 Mark]

#Q. Consider the following two statements:

S_1: Destination MAC address of an ARP reply is a broadcast address. FALSE

S_2: Destination MAC address of an ARP request is a broadcast address. TRUE

Which one of the following choices is correct ?

A. Both S_1 and S_2 are true.

B. S_1 is true and S_2 is false.

C. S_1 is false and S_2 is true.

D. Both S_1 and S_2 are false.

$\boxed{\text{Ans: C}}$

[MCQ] IIT-M, H.W. [GATE-2019][2 Mark]

#Q. Suppose that in an IP-over-Ethernet network, a machine X wishes to find the MAC address of another machine Y in its subnet. Which one of the following techniques can be used for this?

(A) X sends an ARP request packet with broadcast IP address in its local subnet

(B) X sends an ARP request packet to the local gateway's MAC address which then finds the MAC address of Y and sends to X

(C) X sends an ARP request packet with broadcast MAC address in its local subnet

(D) X sends an ARP request packet to the local gateway's IP address which then finds the MAC address of Y and sends to X

Topic : ARP Packet

Frame

Destination MAC Address	Source MAC Address	Type (2 Byte)	Payload [ARP Packet] (28 bytes)	FCS [CRC] (4 Byte)

Header

Footer

Hardware Type (2 Byte)		Protocol Type (2 byte)	
Hardware Length	Protocol Length	Operation (2 byte)	
Source MAC Address (6 byte)			
Source IP Address (4 byte)			
Destination MAC Address (6 byte)			
Destination IP Address (4 byte)			

Topic : ARP Table /ARP cache

→ Used to keep the reply of ARP queries for some time

Host A, ARP Table

IP Address	MAC Address	Time to Live
Host B IP	Host B MAC	30 sec
Router IP	Router MAC	30 sec
Limited B'cast Add	FF:FF... FF	Static

Topic : Loopback Address

→ Know as "localhost"

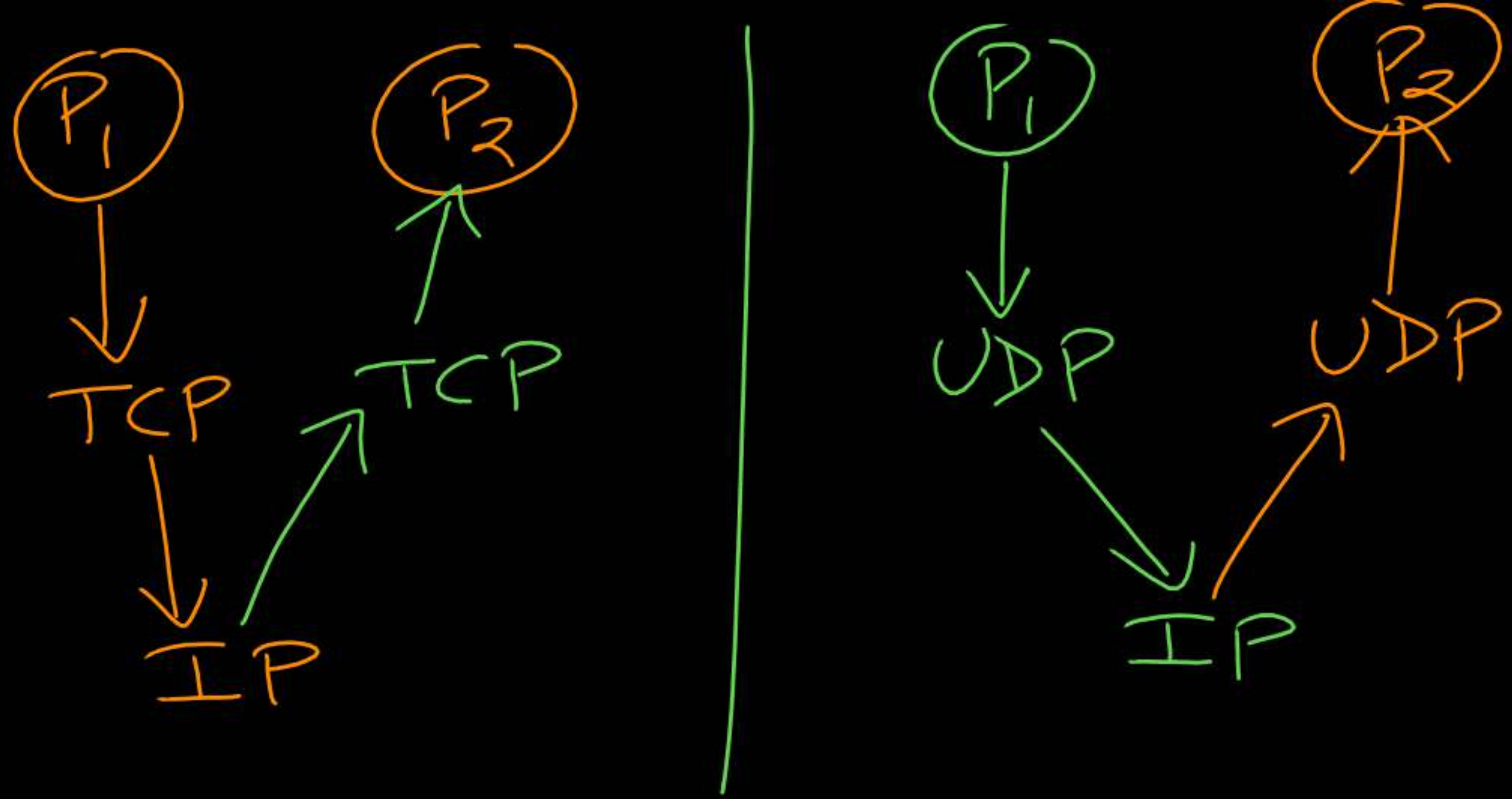
127.Anything

→ In IPv4 loopback address can be anyone from 127.0.0.0 to 127.255.255.255

→ In IPv6 loopback address is "::1"

$\underbrace{00 \ldots 00}_{127 \text{ bit}}1$

→ For diagnostic purpose to verify the internal connectivity


P1
P2
TCP
TCP
IP
P1
P2
UDP
UDP
IP

2 mins Summary

Topic ~~Routing Table~~ Forwarding

ARP

* Supernetting
* NAT
* DHCP

THANK - YOU

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